M. Sample FAT Document

Appendix M of the EFC Gases & Advanced Materials Controls Standard, Revision I, October 14, 2026. Standalone copy; see the full document for the other chapters and appendices.

This appendix is a **sample** Factory Acceptance Test (FAT) document, laid out as a template with a few example rows completed for project `014_R23-Column-PLC`. It is the "FAT Checkout document" of Chapter 3 of the Controls Standard: outlined in Stage 4, procedures and expected results completed by late Stage 5, and actual results recorded in Stage 5A. The structure follows IEC 62381, and the electrical tests follow the testing and verification chapter of NFPA 79.[1] Remove the examples and adapt the sections to each project.

M.1 Cover and Approval

Project no. / name	014 / R23 Column	**System**	Column control panel CP-1
FAT document revision	C	**Test dates**	
PLC software (file)	`014_R23-Column-PLC_C_261002`	**SCADA project**	
Panel builder		**Test location**	

Witness / approval	Name	Signature	Date
EFC Controls			
Panel builder / vendor			
Operations			

M.2 Scope, References, and Test Setup

Scope: the CP-1 control panel, its PLC program, and the SCADA screens for the R23 column, tested at the panel builder's shop before shipment. Field devices are simulated; tests that need the field are listed in Section M.12 for commissioning (SAT).

References (record the revision used): electrical schematics and panel layout; I/O list; Control Narrative; Table of Interlocks and Permissives; alarm list; network map and address list.

[1] IEC 62381:2024 (Ed. 3), *Automation systems in the process industry — Factory acceptance test (FAT), site acceptance test (SAT), and site integration test (SIT)*; NFPA 79-2024, Ch. 18 (18.2 continuity of the effective ground-fault current path, 18.3 insulation resistance, 18.4 voltage tests, 18.6 functional tests); IEC 60204-1:2016+A1:2021, Cl. 18; UL 508A, 3rd ed. (panel construction and markings).

Test setup: how field signals are simulated (loop calibrator, switches, PLC simulation), the SCADA computer used, and the test equipment with calibration dates (multimeter, insulation tester, loop calibrator).

M.3 Entry Criteria

☐ Drawings at the revision under test

☐ PLC and SCADA software revision recorded

☐ Test equipment calibrated

☐ Builder's own pre-FAT checks complete and signed

☐ Test procedures reviewed and approved

☐ Open items from earlier reviews listed

M.4 Documentation and Visual Inspection

Table M.1: FAT — Documentation and Visual Inspection

#	Check	Result / remarks	P/F	Init.
V1	Nameplate: voltage, phase, frequency, FLA, SCCR, enclosure type, drawing number			
V2	UL label present (if a listed panel is required)			
V3	Components match the bill of materials and panel layout			
V4	Wire numbers and terminal markers at both ends, per schematic			
V5	Wire colors per the project scheme; externally powered wiring identified			
V6	Bonding: back panel, door jumper, DIN rails, ground bar (Section 6.3.5 of the Controls Standard)			
V7	High-voltage separation, barriers, and warning labels			
V8	Spare space and spare terminals as specified (at least 20%)			
V9	Workmanship: duct fill, strain relief, no damaged insulation, torque marks			

M.5 Electrical Tests

Table M.2: FAT — Electrical Tests

#	Test	Acceptance	Measured	P/F	Init.
E1	Continuity of the protective bonding circuit from the PE terminal to each exposed metal part	Low resistance per NFPA 79 18.2			
E2	Insulation resistance, power circuits to PE (electronics disconnected)	$\geq 1\,\text{M}\Omega$ at 500 VDC			
E3	Power-up: incoming voltage, control transformer secondary, 24 VDC supplies	Within ±5% of nominal	*Example:* 24.1 VDC	*P*	*AZ*

#	Test	Acceptance	Measured	P/F	Init.
E4	UPS (if fitted): transfer on loss of input	PLC and network stay powered			
E5	Fuses and breakers: ratings match the schematic	As drawn			

M.6 I/O Point-to-Point Checks

Each point is checked from the terminal (simulated field signal) through the PLC to the SCADA screen. Analog inputs are checked at 0, 25, 50, 75, and 100% of range.

Table M.3: FAT — I/O Checks

Tag	Description	Card / ch.	Simulated / expected	PLC / SCADA reading	P/F	Init.
ZSC-110	*V-110 closed limit (DI)*	*A02 / 4*	*Switch closed → 1*	*1 / Closed*	*P*	*AZ*
SV-110	*V-110 solenoid (DO)*	*A04 / 2*	*Command on → 24 V at TB3-12*	*24.0 V*	*P*	*AZ*
PT-210	*Vacuum header (AI, 0–100 torr)*	*A06 / 3*	*4 / 8 / 12 / 16 / 20 mA → 0 / 25 / 50 / 75 / 100 torr*	*0.1 / 25.0 / 50.0 / 75.1 / 99.9*	*P*	*AZ*
FCV-120	*Feed valve (AO)*	*A07 / 1*	*0 / 50 / 100% → 4 / 12 / 20 mA*			

M.7 Communications and Network

☐ IP addresses match the address list
☐ SCADA connects over OPC UA with security enabled
☐ Loss of a device: alarm and fail-safe action
☐ All EtherNet/IP devices connected and healthy
☐ NAT mappings and firewall rules as designed
☐ Loss of SCADA: PLC continues to control safely

M.8 E-Stop and Safety Circuit Tests

Table M.4: FAT — E-Stop and Safety Circuits

#	Test (Section 6.3.3 of the Controls Standard)	Expected result	P/F	Init.
S1	Press each E-stop in turn	Safety relay drops out; outputs and STO removed; E-stop lights; PLC/SCADA show which E-stop		
S2	Open each channel separately (jumper removed)	Safety relay trips and will not reset (cross-fault / channel fault detected)		
S3	Release E-stop; observe reset light	Light blinks (reset required); no restart		
S4	Press and release reset	Safety relay resets; light steady; equipment does not start by itself		
S5	Hold reset button pressed during E-stop release	No reset until button is released (monitored manual reset)		
S6	PLC drops the force-guided relay output	System trips as for an E-stop		

#	Test	Expected result	P/F	Init.
S7	VFD STO status read by PLC during E-stop	SCADA shows drive in STO		

M.9 Functional Tests From the Control Narrative

Table M.5: FAT — Functional Tests

#	Test	Expected result	P/F	Init.
F1	Mode changes Off / Manual / Auto / Lockout for each device	Mode follows request only when permitted; SCADA shows mode		
F2	Start the fill sequence with all permissives met	Steps run in order per the Control Narrative		
F3	Permissive: set the max vacuum-open pressure setpoint to its minimum and maximum limits, and try a value outside them	Values within limits accepted; outside values clamped; limits shown on SCADA		
F4	Each interlock in the Table of Interlocks and Permissives	Correct action; first-out shows the initiating condition; reset only after the cause clears		
F5	Each alarm in the alarm list	Activates at setpoint after delay; clears with deadband; correct priority and message		
F6	Bypass (if used)	Requires authorization; visible on SCADA; logged		

M.10 Abnormal and Recovery Tests

☐ Power loss and restore: no unexpected start
☐ Analog bad quality (open loop, <3.6 mA)
☐ Download/restart: sequences restart as designed
☐ PLC stop/run: outputs to defined state
☐ SCADA disconnect and reconnect
☐ Device communication loss

M.11 SCADA/HMI Tests

☐ Navigation and every screen opens
☐ Faceplates match the device data types
☐ Trends and history recording
☐ Screen scaling at deployed resolutions (Appendix A of the Controls Standard)
☐ Security roles: operator vs. engineer rights
☐ Alarm display, acknowledge, shelving

M.12 Punch List and Field Tests

Table M.6: FAT — Punch List

#	Item	Severity	Owner	Due	Closed
P1	*PT-210 faceplate shows wrong units (psig instead of torr)*	*Minor*	*Controls*	*Before shipment*	
P2	*Field test: verify V-110 stroke time (needs air at site)*	*Field*	*Controls*	*Stage 6*	

Severity: *Critical* (must be fixed before shipment), *Minor* (may ship; fix by a date), *Field* (can only be tested on site).

M.13 Final Acceptance

☐ All tests passed or recorded on the punch list ☐ No open critical items ☐ Software revision and backup recorded

FAT accepted by	Name	Signature	Date
EFC Controls			
Panel builder / vendor			